

INTRODUCTION

Inventory comprises of usable but idle assets. The assets may be of any type-men, material machine etc. For example, if a firm buying a machine, employ a professional in eagerness of the necessities of their facility in future, these source labors as inventory. Usually inventory of men or machines is not accepted and organizations hire equipment of consult professional whenever required, as an cost-effective substitute. When the resource involved is material or goods. It is mentioned as stock or merely as 'Inventory'. The item is generally used specify raw materials in route, finished product, packaging, spares and others-stocked in direction to meet an predictable demand or distribution in the future. Though inventory of materials is an idle assets – it is not meant for direct use—it is nearly important to sustain some inventories for the smooth functioning of an arrive expense.

CHAPTER I Discuss the elementary terms of inventory such as cost and factors of inventory have been discussed.

CHAPTER II Basic notations and assumptions were given.
Mathematical model have been developed.

CHAPTER III Comprise of numerical examples.

CHAPTER – I

BASIC DEFINITIONS

1.1. INTRODUCTION TO OR:

The British/Europeans discuss to “operation research” the Americans to operation research but both are commonly condensed to just “OR” which is the word we will use.

Another word which is used for this field is “management science” (“MS”). The Americans occasionally combine the term OR and MS organized and say “OR/MS” or “ORMS”. However other term occasionally used for “industrial engineering” (“IE”) and “decision science” (“DS”). In current year there has been a additional to a regulation upon a particular term for the field, label it the word “OR”

Operation research moderately is an innovative discipline. The matters and the limitations of the OR are not yet stable. Therefore, to give a proper definition of the word Operation research is a hard job. The OR starts when mathematical and quantitative methods are used to demonstrate the result being engaged. The main action of a administrator is the decision building. In our routine life we make the choices even without notice them. The decision are taken basically by public logic, judgment and expertise without consuming any measured or additional model in simple state but the choice we are concerned here are multifaceted and seriously accountable. Examples the public shipping network design in a city heaving its own design of factories, residential blocks or finding the suitable products mix when there be a huge number of products with dissimilar turnover contributions and production requirements etc.,

Operation research tools are not from any unique discipline. Operation research take tools from dissimilar discipline such as Mathematics, Economics, Psychology, set of facts for decision making. Nowadays OR become a professional discipline, and specifically to the distribution of scare resources. The main resolution of OR is to afford a rational base for decisions

building in the lack of whole facts, because of the structure composed of human, machine and techniques may do not have thorough evidence.

Operation research can also treated as science in the logic defining accepting and expecting the system behavior man machine organization. Thus OR experts are involved in there classical feature of science, they are as follows:

- 1) Deteriorating the organization behavior
- 2) Analyzing the system behavior by emerging suitable model
- 3) Forecast the upcoming behavior by these models

The importance on analysis operation as a whole differentiates the OR from other research and engineering OR is an interdisciplinary discipline which offer solutions to difficulties of military operations during World War 2, and also effective in other operations. Currently business applications are predominantly concerned with OR analysis for the substitute movements. The business and industry suited from OR. In the regions of inventory, reorder policies, optimum position and size of warehouses, adverting, policies, etc.,

1.2. STAGES OF DEVELOPMENT OF OPERATON RESEARCH:

OR are also known as segments and progression of OR which has six significant steps. These steps are organized in the following order.

Step 1:

Perceive the problem environment.

Step 2:

Analyze and describe the problem.

Step 3:

Improve the model.

Step 4:

Select suitable data input.

Step 5:

Afford a solution and check its reasonableness.

Step 6:

Implement the solution.

Step 1: Perceive the problem environment.

The first step in the procedure of OR developed is the problem environment observation. This step comprises diverse events they are conference, site visit research observation etc. These events offer enough information to the OR specialist to formulated the problem.

Step 2: Analyze and describe the problem.

In this step in addition to the problem definition the goals uses the limitations of OR. Study of the problem also well-defined. The output of these step are perfect collection of need for a solution and its nature understanding.

Step 3: Improve the model.

A model is depictions of some abstract or actual situation. The models are essentially mathematical models which defines systems, process in the form of equation, relationships. The diverse activities in this step are variables definitions, formulating, equation etc. The model is verified in this field under diverse environmental constraints and improved in order to work. Earlier the model is improved to satisfy the management with the result.

Step 4: Select suitable data input.

A model work aptly when there is proper data input. Hence selecting proper input data is significant step in the OR. Develop phases or process. The activities in this step comprise internal/external data analysis, fact analysis and collection of options and usage of computer data banks. The objective of this step is to offer adequate data input to activate and test the model developed in step 3.

Step 5: Afford a solution and check its reasonableness.

This solution is not executed instantaneously instead the solution is used to check the model and to find there are any restrictions. Suppose the solution is not sensible or the behavior of the model is not appropriate, the model is rationalized and modified at these stages. The out of this stage is the results that support the existing organizational intentions.

Step 6: Implement the solution.

At this step the solution attained from the precious step is applied. The implementation of the solution includes more many behavior issues. Therefore, before executions the authority has to resolve the matters. A properly realized solution results in worth of work and increases the support from the organization.

1.3. APPLICATIONS OF OPERATION RESEARCH:

All fields of business and government consuming the profits of operation research .There are voluminous of applications of operation research. Accounting, Constructions, Facilities, Planning, Finance, Manufacturing, Marketing Organizational behavior/human resources, Purchasing, Research and Development.

1.4. WHY WE NEED INVENTORY CONTROL:

Every production department ie., a factory, a workshop or an engineering department has to handle materials are of three fold:

1. Increase the customer's service
2. Decrease the investment on materials
3. Co-efficient on low cost plant operation

The above three objectives are based in conflict and need for particular purpose to get an optimal solution with a view to earn increase gain for a given investment study of the connected operation are now known by the term **“INVENTORY CONTROL”**.

1.5. SOME BASIC DEFINITION:

1.5.1. Demand

The number of units considered per period is called **“Demand”**.

1.5.2. Lead time

The lead time between placing of an order and its actual arrival in the inventory is known as **“Lead Time”**.

1.5.3. Cycle time

The time period between placements of two successive orders is called **“Cycle Time”**. The order may be two types of inventory review systems.

(a)Continuous review:

“The data of the inventory will be continuously scrutinized up to the new order”.

(b) Periodic review:

“This system the inventory levels are observed at equal time intervals and orders are placed at such intervals”.

1.5.4. Time horizon:

The time period over which the inventory level be limited is called the **“Time Horizon”**.

1.5.5. Reorder level:

“The level between higher and lower stock, at which purchasing activities must start for supply”.

1.5.6. Order quantity:

“The quantity of material ordered each time inventory is supplied. Q is stable in fixed order quantity systems and unstable in fixed order period systems”.

1.5.7. Stock Replenishment:

“The process of including material to inventory. It contains sensing the need for an order of materials, placing the order, production of the order conveyance and receipt of the order”.

1.5.8. Backlogging:

“The process of holding customer order to be filled later when they cannot be shipped instantly because of stock outs”.

1.5.9. Back order:

“An item ordered but cannot be delivered on the which the customer will for delivery obscene dropping the order”.

1.5.10. Stock out:

“When the utilizable inventory level for a material is decreased to zero. A situation when an order is placed for an piece that is not on hand in inventory”.

1.5.11. Inventory Turnover:

“**Inventory turnover**” is defined as the ratio of the value of material consumed to the moderate investment in inventories for the similar period.

$$\text{Inventory turnover} = \frac{\text{value of the material consumed}}{\text{value of the average inventory}}$$

1.5.12. Standardisation:

“**Standardisation**” is the determination of standard sizes, shapes, quantity and aspect of material.

1.5.13. Demand rate:

“The number of units demanded by purchasers of production departments per unit of time. Need to be in the similar units as the “**supply rate**”.

1.5.14. Supply rate:

“The number of units per unit of time of an materials supplied to inventory if delivered moderately. Must be in the same units as “**demand rate**”.

1.6. COSTS ASSOCIATED WITH INVENTORIES:

The elementary concepts that are associated to inventory are discussed as follows.

1.6.1. Purchase cost:

The cost of purchasing a unit of item. The “**purchase price**” will become important when quantity discounts are allowed for ‘purchase’ above a certain quantity.

$$\text{Purchase cost} = \text{price per unit} * \text{demand per unit}$$

1.6.2. Carrying cost or holding cost:

These are the costs incurred for shifting or belongings inventory items in the storeroom. They include,

- Warehouse cost for providing storeroom space to save the products.
- Inventory holding cost for the payment of payments to worker.
- Security cost adverse possible loss from fire or other form of damage.

1.6.3. Ordering cost:

“**Ordering cost**” adding all cost when a demand is placed for purchasing item from the external dealers. The cost per order basically includes

- Addition cost of handling of invoices consistent, payments etc..
- Cost of services which embrace cost of mailing, telephone calls, sequestration and other follow up actions
- Material handling cost suffered in receiving, sorting, searching and keeping the items included in the order.
- Account the audit, etc.,

1.6.4. Shortage or stock out cost:

The sanction cost for running out of stock (i.e when an item unable supplied on the customer's demand) is known as "**shortage cost**". This cost includes the loss of probable gain through sales of piece and loss of stable loss of customers and its associated lost gain in future sales.

1.6.5. Salvage cost (or selling price):

When the demand for certain material is affected by the quantity stocked, decision problem is norm n a profit maximization basis that includes the income from "**selling**".

1.6.6. Set -up cost:

This the cost integrated with the "**setting up**" of machinery before starting the production and this cost is moderately individualistic of the quantity ordered/produced.

1.7. MAINTAINING THE INVENTORY IS NECESSARY BECAUSE OF THE FOLLOWING REASON:

- Maintaining the inventory helps smooth and effective running of enter price. It decouple the production from customers and supplier and simplifies the complex organization for production and decreases the co-ordinations etc.
- It provides service to the customer at a tiny notice. Timely deliveries can fetch more good will and orders.
- It may earn reduction because of massive purchasing.
- An agency may have to deal with many customers and vendors who are not surely near it. Therefore have to be built to meet the demand at least during the transfer time.
- Maintaining of inventories helps in maintaining economy by interesting some of the fluctuations When the pieces and for an pieces fluctuates and seasonal.

1.8. THE CONCEPT OF EOQ:

In the inventory problem demand is assumed to be standard and completely pre-determined are traditionally to as the economic order quantity (EOQ) or lot size problem.

By the order quantity we mean the quantity produced or procured during one production cycle. When the size of order maximum, the ordering cost (cost of purchasing, inspection etc.) will minimize where as the inventory carrying cost (cost of storage, insurance, etc.) will increase. Thus in the production task there are two opposite costs, one encourage the increase in the order size and the other discourages, economic order quantity (EOQ) is that the size of order which decreases total annual costs of carrying inventory and cost of ordering.

CHAPTER –II

AN INVENTORY MODEL FOR DETERIORATING PRODUCT UNDER PRESERVATION TECHNOLOGY WITH TIME – DEPENDENT QUALITY DEMAND

2.1 INTRODUCTION

Inventory management is an essential aspect of the trade. The objective of an inventory of an inventory problem is to reduce the total cost of the inventory system by making the best decisions based on the mathematical methods that can be used to solve problems, such as optimum control analysis, dynamic programming and network optimization. Generally, inventory decisions revolve around two questions: when should the inventory be renewed and how much order should be added to the inventory?

Inventory is being researched to see if it might help businesses save a lot of money. It is common practice in many firms to store inventory for future sale or use. To meet the demand, firms must have a supply of products on hand that can be sale quickly. The inventory theory objective is to identify the rules that management can use to reduce inventory costs and meet the demand from the market. To achieve an ideal total cost in an inventory model; most of the industrial organizations focus on inventory control and overcoming inventory challenges such as managing and maintaining of degrading items.

Hasan (Hasan et al., 2020) found an imperfect production method for deteriorating agricultural products. The demand for a particular product varies according to the demand rate depending on stock availability, price of items, etc. Sindhuja et al., (Sindhuja et al., 2021) proposed a mathematical model based an quadratic demand and also considered time-dependent demand. Few research works which focuses on both demand and deterioration. An inventory is decreased by demand was developed by Bragila et al., (Bragila et al., 2019).

Perishable products have high and varying deterioration rates when passing through their logistics system. Few recent technical advancements includes preservation technology, supply chain technology and warehouse technology that have resulted in the

development of new technologies that can be utilized to effectively measures and regulate the deterioration. In practice, the deterioration rate of products can be controlled and reduced through various efforts such as operational changes and specialized equipment acquisition. By the efforts of investing in reducing the deterioration rate, the supplier can reduce the economic losses, improve the services level to the retailer, and increase business competitiveness.

Preservation technology is an essential component in minimizing the deteriorating effect. It may be used to measure and control the pace of deterioration effectively. In practice, the deterioration rate of products can be controlled and reduced through various efforts such as operational changes and specialized equipment acquisition. The supplier can reduce economic losses, improve the service level to the retailer, and increase business competitiveness by investing in preservation technology and lowering the deterioration rate.

In this context, several businesses and organization are required to use preservation technology in their inventory management system. Hsieh and Dye (Hsieh & Dye,2013) derived a production model under time-dependent demand with a controllable deterioration effort. A deterioration inventory model with preservation technology under stock dependent demand was developed by Zhang et al.(Zhang & Tang, 2014). Yang et al.(Yang et al, 2015) proposed a preservation inventory model under trade credit with a deterioration effect. A preservation inventory model under price-dependent demand with short-age was proposed by Mishra et al. (Mishra et al., 2017). Suphash Chandra Das et al. (Das et al., 2020) proposed a mathematical model based on an application of preservation technology which is followed by selling price dependent demand. Khanna et al. proposed a model based on time-varying holding cost under the investment of preservation technology. Carbon control policies by developing an optimal inventory model dealing with deteriorating items with preservation technology are developed by Khanna et al. Sustainable production strategic for deteriorating and imperfect quality items with an investment in preservation technology are

presented by Priyamvada et al. Das et al. proposed an inventory model based on preservation technology for a non-instantaneous deteriorated inventory model considering multi-period trade credit policy and variable demand dependent on selling price. Mishra et al. (Mishra et al., 2017) proposed a model based on the preservation technology, this paper again revisited by Priyamvada et al.

2.2 Research gap and contributions

In this project, an attempt has been made to develop an inventory model with a controllable deterioration rate by allowing preservation technology parameter as the most important decision variable in conjunction with the inventory system. The quality control deterioration rate by choosing the cost containment with preservation effort. The main objective of this model is to find the optimal time cycle length and preservation technology investment strategies while minimizing the total cost per unit time.

Few research paper based on prevention technology are identified and compared proposed model. The report focuses on what sets the study apart from the literature: Introducing the holding cost with controllable prevention and constant holding cost demand is related to the quality of the perishable commodity, which is determined by its rate of deterioration. The effort of the deterioration on items, as well as prevention technology measures to control the rate of deterioration. By applying the model, the supplier benefits at each point, decreasing the order quantity to decrease the production loss to deterioration. Furthermore, the practitioner must choose not only the traditional order quantity decision, but also the quantity control deterioration rate by choosing the following factors.

- 1) The depreciating worth if the goods
- 2) Cost containment with preservation effort
- 3) Quality-sensitive demand

2.3 Description, notations and assumptions

In this section, the model's description is explained and the components are elaborated under notations, and assumptions are mentioned below as well.

Problem description

The policy of a dairy company, in particular, an ice cream factory, should be set so that the average overall cost is kept as low as possible. The perishable nature of materials presents significant challenge, putting constraints on planning systems. The absence of freshness, especially noticeable in ice cream maintained in a stable environment, is one of the most considerable factors determining the pace of degradation. Additionally, ice cream can absorb volatile flavours from the storage environment. The cost of using modern preservation technologies, on the other hand, has been estimated because of the suggested model, the supplier benefits by determining an optimal time cycle duration strategy.

Notations

In this inventory management problem, the following notations are utilized

- A -represents the fixed cost of ordering.
- Θ -is the deterioration rate.
- ε -is the preservation indicator.
- h -is the constant holding cost.
- h_{ϕ} -is the actual controlled holding cost when it comes to preservation.
- h_{α} -is the maximum preservation holding cost per unit and per time unit.
- TC -is the overall cost.
- The demand D_d is related to the quality of the perishable commodity, which is influenced by its rate of deterioration.
- d is fixed demand.
- β -demand-on-quality variable.
- C is the deterioration cost value per unit.
- s is the shortage cost per unit.

Assumptions

The mathematical model proposed in this project based on the following assumptions.

- A single item is considered in this inventory system.
- A deterministic demand pattern is considered.
- The demand is dependent on the product quality. The quality of the perishable commodity, which in turn depends on its rate of deterioration, determines demand.
- Ordering cost is unchangeable. Assume that an order containing a specific type of preserved product has a higher ordering cost. In general, the initial freshness of each and every product ordered determines the retailer's maximum effective preservation effort. The preservation benefits include the expense of adopting advanced preservation technologies that have been pre-allocated.
- The holding cost with preservation effort is directly related to the expected deterioration by allocating required personal costs, equipment and other operational expenditures.
- The rate of deterioration is quite low when a corporation makes effort to preserve a product as $h \rightarrow h_\alpha$. In this case, the highest level of product quality is maintained. Because preservation necessitates the use of storage devices, preservation equipment, higher power efficiency, and human labour, the holding cost linked to the quality indicator.
- Shortages are allowed.
- Deteriorated items should not be replaced during the cycle period.
- Lead time is assumed to be zero.

2.4. Existing model

Subhash Chandra Das et al. (Das et al., 2020) proposed a mathematical model based on an application of preservation technology and selling price-dependent demand. Considering the instantaneous inventory level $I(t)$ at any time t is determined by the following differential equations:

$$\frac{d I(t)}{dt} = -D(P), 0 \leq t \leq \mu \quad (1)$$

$$\frac{d I(t)}{dt} = -D(P) - \theta I(t)(1 - m\mu), 0 \leq t \leq t_p \quad (2)$$

From the above equations, it is observed that $I(t)$ is the inventory amount at any time “t” which is defined by the differential equations based on demand and deterioration .

The deterioration $\theta(t)$ is considered by three-parameter Weibull distribution deterioration rate,

$$\text{i.e., } \theta(t) = \alpha\beta(t - \gamma)^{\beta-1}$$

2.5 Proposed mathematical model :

In this section, a mathematical model is developed to minimize the total inventory cost under the condition of preservation technology with quality-dependent demand on time. The proposed model can be defined by the inventory level decreases due to demand and degradation of the goods over the period $[0; T_1]$, according to the notations and assumptions indicated above. When the inventory level goes to zero at the time $T = T_1$ the shortages begin and the inventory level continues throughout the period from T_1 to T .

Considering the effect of the quality demand and the condition of preservation technology, under various conditions of deterioration rate, the order quantity varies. Each order cycle results in a decrease in the inventory level due to both demand and product deterioration. The inventory level at any point of time t , $I(t)$ can be expressed by the following differential equation:

$$\frac{dI(t)}{dt} + \Theta(1-\xi) I(t) = -D_d(t), 0 \leq t \leq T_1 \quad (3)$$

$$D_d = (d - \beta\Theta)$$

The boundary conditions are $I(0) = q_1$ and $I(T_1) = 0$. Solving the equation (3) the inventory level can be expressed as

$$\frac{dI(t)}{dt} = -D_d(t), T_1 \leq t \leq T \quad (4)$$

In equation 3 by applying the boundary condition $I(T_1)$, $I(T)$ is obtained as below

$$I(t) = \frac{e^{-T_1(-\theta+\theta\xi)}(e^{t(\theta+\theta\xi)} - e^{T_1(-\theta+\theta\xi)})(-d+\beta\theta)}{\theta(-1+\xi)} \quad (5)$$

To apply $t = 0$ in the preceding equation, use $I(0) = q_1$ as a condition, hence q_1 is represented by

$$q_1 = \frac{e^{-T_1(-\theta+\theta\xi)}(1 - e^{T_1(-\theta+\theta\xi)})(-d+\beta\theta)}{\theta(-1+\xi)} \quad (6)$$

The solution of equation (4) is yielded below by applying the condition $I(T_1) = 0$ is

$$q_2 = (T + T_1)(d - \beta\theta) \quad (7)$$

Initial order quantity is $Q_i = q_1 + q_2$

$$Q_i = \frac{(\beta\theta - d)e^{(\xi-1)\theta(-T_1)}(1 - e^{(\xi-1)\theta(-T_1)})}{(\xi-1)\theta} \quad (8)$$

Ordering cost "A"—makes up the overall relevant cost of each cycle.

Inventory holding cost in the system during the time intervals $[0, T_1]$ and the perishable inventory.

- The constant holding cost h is comprises both physical storage and opportunity costs.
- h_α - is the cost of freshness preservation, which is incurred to keep the storage state under control.

Holding cost $C_{hi} = (h + h_\alpha(1 - \Theta)^\xi) \int_0^{T_1} I(t) dt$

$$= \frac{e^{-T_1}(e^{-T_1} + e^T(-1+T-T_1))(d-\beta\theta)(\theta-\theta\xi)^2}{\theta(-1+\xi)} \quad (9)$$

The number of units deteriorated is $Q = Q_i$. Hence the deterioration cost is

$$D_r = c \left(Q_i \int_0^{T_1} (d - \beta\theta) dt \right) \\ = c \left(\frac{(\beta\theta - d) e^{(\xi-1)\theta(-T_1)} (1 - e^{(\xi-1)\theta(-T_1)})}{((\xi-1)\theta)} - T_1 (d - \beta\theta) \right) \quad (10)$$

During a cycle, the loss caused by a shortage, the stockout cost between the time intervals (T_1, T) is given by

$$S_c = -s \frac{1}{2} (T - T_1)^2 (d - \beta\theta) \quad (11)$$

Given the limitations of preservation technology, the preservation technology investment cost is calculated as follows:

$$\text{Preservation cost } P_c = \xi T \quad (12)$$

The total cost per cycle z is $\frac{1}{T} [A + C_{hi} + S_c + D_r + P_c]$

$$z = \frac{1}{T} \left[A + (h + h_\alpha (1 - \theta)^\xi) \left(\frac{e^{T_1} (e^{T_1} + e^T (-1 + T - T_1)) (d - \beta\theta) (\theta - \theta\xi)^2}{\theta(-1 + \xi)} \right) \right] \\ + c \left(\frac{(\beta\theta - d) e^{(\xi-1)\theta(-T_1)} (1 - e^{(\xi-1)\theta(-T_1)})}{((\xi-1)\theta)} - T_1 (d - \beta\theta) \right) - \\ s \frac{1}{2} (T - T_1)^2 (d - \beta\theta) + \xi T \quad (13)$$

To resolve the approximation exponential function, equation (13) is obtained.

$$z = \frac{1}{T} \left[A + (h + h_\alpha (1 - \theta)^\xi) \left(\frac{-4 (T - T_1) (d - \beta\theta) (\theta - \theta\xi)^2}{\theta(-1 + \xi)} \right) \right] + \\ c \left(\frac{(d - \beta\theta) T_1 (-1 + \xi)}{((\xi-1)\theta)} - T_1 (d - \beta\theta) \right) - \\ s \frac{1}{2} (T - T_1)^2 (d - \beta\theta) + \xi T \quad (14)$$

The total relevant cost can be reduced based on the following necessary conditions.

$$\frac{\partial(TC(T,T_1))}{\partial T}=0 \text{ and } \frac{\partial(TC(T,T_1))}{\partial T_1}=0$$

The optimal values of $T=T^*$ and $T_1=T_1^*$ are obtained by simultaneously solving.

$$\frac{\partial(TC(T,T_1))}{\partial T}=0 \text{ and } \frac{\partial(TC(T,T_1))}{\partial T_1}=0$$

2.6 Solution approach :

Total cost analysis based on deduction of optimal cost in equation (14) is analyzed below.

2.6.1. Total cost analysis based on deduction optimal cost

According to the optimality necessary condition, derive equation (14) with respect to T and T_1 .

$$\begin{aligned} \frac{\partial(TC(T,T_1))}{\partial T} = & -\frac{1}{T} \left[A + (h + h_\alpha(1 - \theta)^\xi) \left(\frac{-4(T-T_1)(d-\beta\theta)(\theta-\theta\xi)^2}{\theta(-1+\xi)} \right) \right] \\ & + \\ & c \left(\frac{(d-\beta\theta)T_1(-1+\xi)}{((\xi-1)\theta)} - T_1(d-\beta\theta) \right) - s \frac{1}{2} (T-T_1)^2(d-\beta\theta) + \\ & \xi T \end{aligned} \quad (15)$$

$$\begin{aligned} \frac{\partial(TC(T,T_1))}{\partial T_1} = & c - d + \beta\theta + s(T-T_1)(d-\beta\theta) + \\ & \frac{(4(d-\beta\theta)(h+h_\alpha(1-\theta)^\xi)(\theta-\theta\xi)^2(d-\beta\theta))}{\theta} \end{aligned} \quad (16)$$

$$\frac{\partial^2(TC(T,T_1))}{\partial T_1^2} \left(\frac{\partial^2(TC(T,T_1))}{\partial T^2} \right) - \left(\frac{\partial^2(TC(T,T_1))}{\partial T \partial T_1} \right) > 0 \quad (17)$$

$$\left(\frac{\partial^2(TC(T,T_1))}{\partial T_1^2} \right) > 0 \quad \left(\frac{\partial^2(TC(T,T_1))}{\partial T^2} \right) > 0$$

The optimal values of $T=T^*$ and $T_1=T_1^*$ are obtained by solving simultaneously for

$$\frac{\partial(TC(T,T_1))}{\partial T} = 0 \quad \text{and} \quad \frac{\partial(TC(T,T_1))}{\partial T_1} = 0$$

$$\text{Since} \quad \frac{\partial(TC(T,T_1))}{\partial T} = 0$$

$$(c-d)\Theta + s(T-T_1)(D-\beta\Theta)(\Theta) + \beta\Theta^2 + 4((d-\beta\Theta)(h+h_\alpha(1-\Theta)^\xi(\Theta-\Theta\xi)^2(-1+\xi))) = 0$$

(18)

$$\text{Since} \quad \frac{\partial(TC(T,T_1))}{\partial T} = 0$$

$$\begin{aligned} & 2A\Theta(-1+\xi) + 2(h+h_\alpha(1-\Theta)^\xi(\Theta-\Theta\xi)^2 - 4(T-T_1)(d-\beta\Theta)) + \\ & 2\Theta(d-\beta\Theta) + T_1(-1+\xi) - T_1(d-\beta\Theta)(-1+\xi) \\ & s(T-T_1)(d-\beta\Theta)(\Theta)((-1+\xi)(T-T_1)) + 2\Theta(-1+\xi)(\xi T) = 0 \end{aligned} \quad (19)$$

2.6.2 Solution procedure :

In this section an approach to evaluate the inventory strategy that minimizes the overall cost

per unit time has been described. In an inventory mathematical model, the essential things the ice cream supplier should consider are arranging downtime and ordering up time.

- The life span of the item is assumed as constant deterioration followed by a preservation factor,

$$\text{i.e., } \frac{dI(t)}{dt} + \Theta(1-\xi)I(t) = -(d-\beta\Theta), t, 0 \leq t \leq T_1.$$

- Differential equations are used to calculate inventory's real-time status. The initial condition and boundary condition are used to determine the inventory level. The inventory level at any point of time t, equation (5) represents I(t). The maximum

stock level obtained from the initial condition is $I(0) = q_1$. The initial order quantity is determined before the cost estimation.

- The supplier policies for various estimated costs are taken into account.
- From equations (9), (10), (11) and (12) the estimated costs such as ordering, holding, deterioration, and shortage costs are obtained.
- From equation (9), the actual controllable holding cost with preservation is represented by h_θ . The holding cost with preservation effort directly results in the expected deterioration rate by allocating required labour cost, equipment, and other operational expenditures.
- For each unit and per time unit, h_α is the maximum holding cost with preservation. The rate of deterioration is relatively low when a company makes every effort to preserve a product, such as $h - h_\alpha$. In this case, the highest level of product quality is maintained. Because preservation necessitates the use of storage devices, preservation equipment, higher power consumption, and human labour, the holding cost is linked to the quality indicator.
- In this model, the total cost is composed per cycle denoted as z which is defined as
$$z = \frac{1}{T} [A + C_{hi} + S_c + D_r + P_c],$$
where A, C_{ih}, C_s, D_r are considered as . ordering cost, holding cost, shortage cost, deterioration cost, and preservation cost.
- To obtain the perishable inventory's optimal policies, the model considers the function of time with the total cost per unit time. Differentiating equation (13) with respect to T_1 and T , the optimal values are obtained. Taking the differentiation for equation (13) with respect to T_1 and T , the optimal values of $T_1 = T_1^*$ and $T = T^*$ are obtained.
- Thus the cost variation differs for each flavour of the ice cream which is considered in the case study.

CHAPTER – III

NUMERICAL EXAMPLE

3.1 EXAMPLE 1

Considering the parametric values of the inventory of model of deteriorating items in proper units $A=12,000, h_\phi = 2000, h_\alpha = 2000,$

$\theta = 0.001, \alpha = 0.001, \varepsilon = 0.002, c = 1500, s = 1200, d = 150.$

$\theta = 0.001, \alpha = 0.001, \varepsilon = 0.002, c = 1500, s = 1200, d = 150.$

SOLUTION:

$$T^* = \frac{\partial(TC(T, T_1))}{\partial T} = \frac{-1}{T} \left[A + (h + h_\alpha (1 - \theta)^\xi) \left(\frac{-4(T - T_1)(d - \beta\theta)(\theta - \theta\xi)^2}{\theta(-1 + \xi)} \right) \right] \\ + c \left(\frac{(d - \beta\theta)T_1(-1 + \xi)}{((\xi - 1)\theta)} - T_1(d - \beta\theta) \right) - s \frac{1}{2}(T - T_1)^2(d - \beta\theta) + \xi T$$

Substituting the given parametric values in the above equation, we obtain

$$T^* = 0.81.$$

$$T_1^* = \frac{\partial(TC(T, T_1))}{\partial T_1} = c - d + \beta\theta + s(T - T_1)(d - \beta\theta) + \frac{(4(d - \beta\theta)(h + h_\alpha(1 - \theta)^\xi)(\theta - \theta\xi)^2(d - \beta\theta))}{\theta}$$

We obtain, $T_1^* = 0.75.$

3.2. EXAMPLE 2

Considering the parametric values of the inventory model of deteriorating items $A=12,800, h_\phi = 2500, h_\alpha = 3600, \theta = 0.001, \alpha = 0.001, \varepsilon = 0.002, c = 1500, s = 1200, d = 150.$

SOLUTION:

$$T^* = \frac{\partial(TC(T, T_1))}{\partial T} = \frac{-1}{T} \left[A + (h + h_\alpha (1 - \theta)^\xi) \left(\frac{-4(T - T_1)(d - \beta\theta)(\theta - \theta\xi)^2}{\theta(-1 + \xi)} \right) \right]$$

$$+ c \left(\frac{(d - \beta\theta)T_1(-1 + \xi)}{((\xi - 1)\theta)} - T_1(d - \beta\theta) \right) - \frac{1}{2}(T - T_1)^2(d - \beta\theta) + \xi T$$

Substituting the given parametric values in the above equation, we obtain

$$T^* = 0.82$$

$$T_1^* = \frac{\partial(TC(T, T_1))}{\partial T_1} = c - d + \beta\theta + s(T - T_1)(d - \beta\theta) + \frac{(4(d - \beta\theta)(h + h_\alpha(1 - \theta)^\xi(\theta - \theta\xi)^2(d - \beta\theta))}{\theta}$$

We obtain, $T_1^* = 0.78$.

3.3 EXAMPLE 3

Considering the parametric values of the inventory model of deteriorating items $A=13,800$, $h_\phi = 2800$, $h_\alpha = 3700$, $\theta = 0.001$, $\alpha = 0.001$, $\varepsilon = 0.002$, $c = 1500$, $s = 1200$, $d = 150$.

SOLUTION:

$$T^* = \frac{\partial(TC(T, T_1))}{\partial T} = \frac{-1}{T} \left[A + (h + h_\alpha (1 - \theta)^\xi) \left(\frac{-4(T - T_1)(d - \beta\theta)(\theta - \theta\xi)^2}{\theta(-1 + \xi)} \right) \right]$$

$$+ c \left(\frac{(d - \beta\theta)T_1(-1 + \xi)}{((\xi - 1)\theta)} - T_1(d - \beta\theta) \right) - \frac{1}{2}(T - T_1)^2(d - \beta\theta) + \xi T$$

Substituting the given parametric values in the above equation, we obtain

$$T^* = 0.87.$$

$$T_1^* = \frac{\partial(TC(T, T_1))}{\partial T_1} = c - d + \beta\theta + s(T - T_1)(d - \beta\theta) + \frac{(4(d - \beta\theta)(h + h_\alpha(1 - \theta)^\xi(\theta - \theta\xi)^2(d - \beta\theta))}{\theta}$$

We obtain, $T_1^* = 0$

CONCLUSION

A mathematical model for deteriorating items based on preservation technology with time dependent quality demand under shortages has been developed in this project. The rate of deterioration impacts dairy food products, which increases with time. The cost functions applied in the proposed model benefit the supplier in deciding the optimum operating plan and total sales. Hence, the preservation technology with time-dependent quality demand can be widely used in analyzing situations that arise in the food processing, particularly dairy products (ice cream). After evaluating the model, this proposed model helps the supplier of the company to reduce total costs in terms of the deterioration rate governed by the deteriorating parameter (ϵ) by reducing order size (or) cycle time..

The company reduces its order quantity to inventory loss due to deterioration at each stage. This indicates that the total cost value and beginning order quantity required values are minimized in all circumstances. The proposed model is determined using the quality demand based on the controllable holding cost (h_ϕ) and maximum holding cost (h_α). The significance of the above two parameters enables the supplier to increase the efficiency of maintaining the initial stock order, assure fresh stock availability in the inventory system, and reduce the percentage of lost stock. Thus, the proposed model helps the distributors recover from equipment failure, ice cream formation, and power factor.

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